

HOW THE VERIFIER CHECKS ECONOMIC SAFETY

1 Model inputs

Reserves, C, L, r, F, T
Fixed transaction schedule
Explicit policy assumptions

2 Ask Z3 for a counterexample

Exact rational constants; polynomial transition constraints
QF_NRA query: $\exists$ feasible execution with $\Pi > 0$
The query tests existence; it does not maximize profit.

SAT: a witness exists

Replay rational witnesses
using exact Fraction arithmetic
Check balances and guards
Confirm net extraction

UNSAT: query excluded

No profitable execution
satisfies this encoding
Safety is conditional on
the model and assumptions

UNKNOWN: unresolved

Timeout or solver limit
provides no decisive result
Record it explicitly
Do not label it safe

3 Independent analytical comparison

SymPy-derived criterion: compare $g(0)$, $g(F)$ and an in-range cash-cap kink $g(f_c)$.
Compare the positivity decision with each core sweep result; optimality comes from this analysis.